3DWEBSOFT INC. FOUNDATION

FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

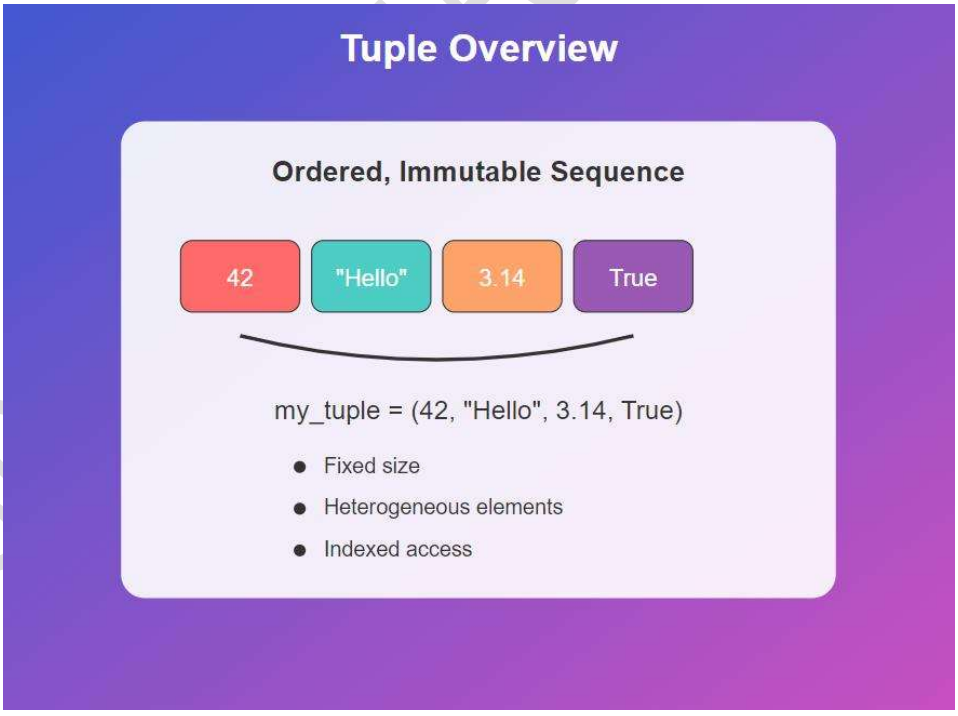
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 28: Tuples (Python)



What is a Tuple?

A **Tuple** is a collection of items that are stored in order.

3D WEBSOFT FOUNDATION

✓ Items are inside **round brackets ( )**.

✓ A tuple **cannot be changed** after it is created.

✓ Tuples can store numbers, words, and different types of data.

Example

fruits = ("Apple", "Banana", "Mango")

print(fruits)

Output

('Apple', 'Banana', 'Mango')

Real Life Example

Imagine your **birthday date**.

Example:

- Day = 25
- Month = August
- Year = 2013

These values never change.

A **Tuple** is like storing permanent information.

Features of Tuple

- Ordered
- Cannot be Changed (Immutable)
- Allows Duplicate Values
- Stores Different Data Types
- Faster than Lists

Tuple Syntax

tuple_name = (item1, item2, item3)

Example

colors = ("Red", "Green", "Blue")

3DWEBSOFT FOUNDATION

Python Tuple Initialization

```
fruits_tuple = ("apple", "banana")
```

```
fruits_tuple = (" 🍎 ", " 🍌 ")
```

Tuple 1 Tuple 2

Tuple Overview

Ordered, Immutable Sequence



my_tuple = (42, "Hello", 3.14, True)

- Fixed size
- Heterogeneous elements
- Indexed access



Tuple

(

Apple

Banana

Mango

)

Creating Tuples

```
animals = ("Dog", "Cat", "Tiger")
print(animals)
```

Output

('Dog', 'Cat', 'Tiger')

Tuple with Numbers

3DWEBSOFT FOU

```
numbers = (10,20,30,40)
print(numbers)
```

Output

(10, 20, 30, 40)

Mixed Tuple

```
student = ("Rahul",12,90.5)
print(student)
```

Output

('Rahul', 12, 90.5)

Accessing Tuple Items

Index starts from **0**

```
fruits = ("Apple","Banana","Mango")

print(fruits[0])
```

Output

Apple

```
print(fruits[1])
```

Output

Banana

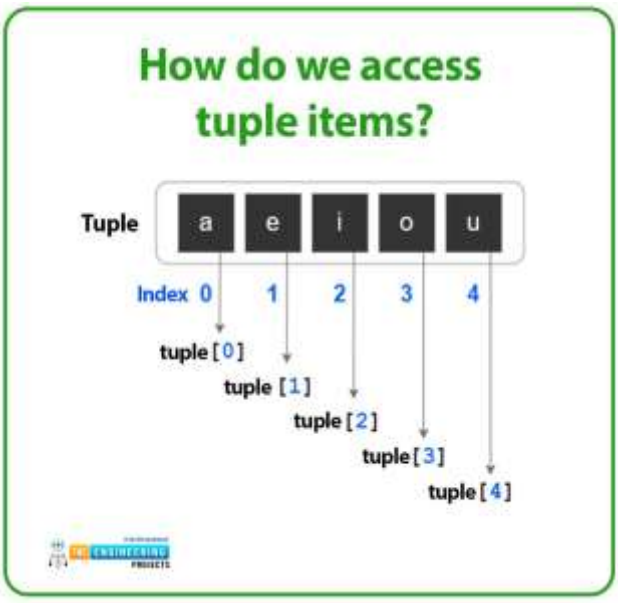
```
print(fruits[2])
```

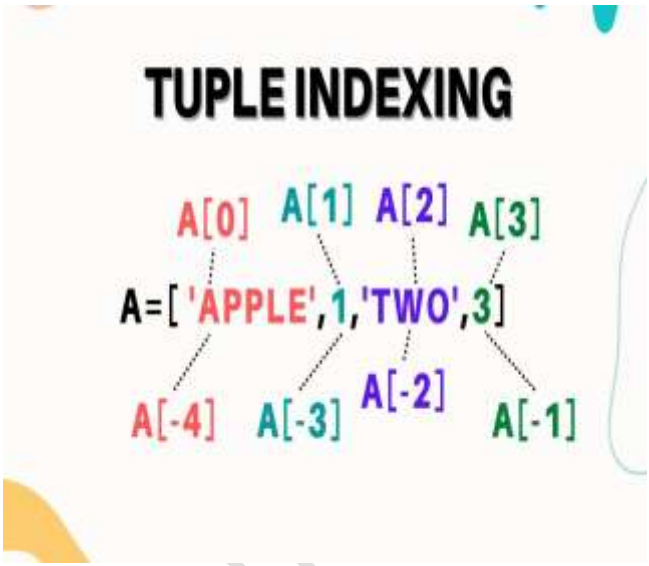
Output

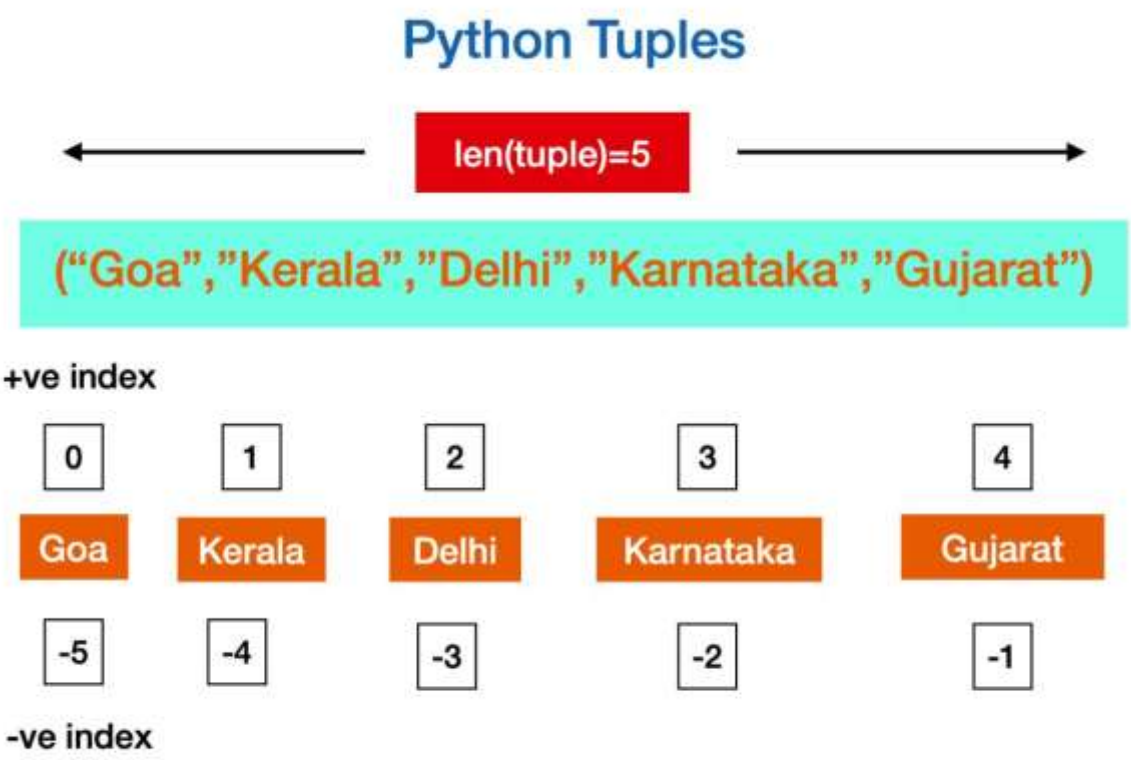
Mango

Tuple Index

3DWEBSOFT FOUNDATION







Tuple

Apple	Banana	Mango
0	1	2

Negative Index

```
fruits = ("Apple","Banana","Mango")

print(fruits[-1])
```

Output

Mango

Tuple Length

```
fruits = ("Apple","Banana","Mango")

print(len(fruits))
```

Output

3

Loop Through Tuple

```
colors = ("Red","Blue","Green")


for color in colors:
   print(color)
```

Output

Red

Blue

Green

Check Item Exists

```
fruits = ("Apple","Banana","Mango")


print("Apple" in fruits)
```

Output

True

3DWEBSOFT FOUNDATION

Tuple Concatenation

```
a = (1,2,3)

b = (4,5)


print(a+b)
```

Output

(1,2,3,4,5)

Tuple Repetition

```
numbers = (1,2)


print(numbers*3)
```

Output

(1,2,1,2,1,2)

Tuple Slicing

```
numbers = (10,20,30,40,50)


print(numbers[1:4])
```

Output

(20,30,40)

Count Function

```
numbers = (1,2,2,3,2)


print(numbers.count(2))
```

Output

3DWEBSOFT FOUNDATION

3

Index Function

```
colors = ("Red","Blue","Green")


print(colors.index("Blue"))
```

Output

1

Convert Tuple to List

```
fruits = ("Apple","Banana","Mango")


mylist = list(fruits)


print(mylist)
```

Output

['Apple', 'Banana', 'Mango']

Convert List to Tuple

```
mylist = ["Apple","Banana","Mango"]


mytuple = tuple(mylist)


print(mytuple)
```

Output

('Apple', 'Banana', 'Mango')

Nested Tuple

3DWEBSOFT FOUNDATION

```
marks = (
(90,80),
(70,60)
)

print(marks)
```

Output

```
((90,80),(70,60))
```

Tuple Packing

```
student = "Rahul",12,"A"

print(student)
```

Output

```
('Rahul',12,'A')
```

Tuple Unpacking

```
student = ("Rahul",12,"A")

name,age,grade = student

print(name)
print(age)
print(grade)
```

Output

```
Rahul
12
A
```

3DWEBSOFT FOUNDATION

❌ **Tuples Cannot be Changed**

```
fruits = ("Apple","Banana","Mango")


fruits[0] = "Orange"
```

Output

TypeError

Because tuples are **immutable**.

Difference Between List and Tuple

List	Tuple
[] Square Brackets	() Round Brackets
Changeable	Not Changeable
Slower	Faster
Mutable	Immutable

Real-Life Examples

Student Details

```
student=("Rahul",13,"8th Class")
```

Days of Weekend

```
weekend=("Saturday","Sunday")
```

Traffic Lights

```
signals=("Red","Yellow","Green")
```

Months

```
months=("Jan","Feb","Mar")
```

3DWEBSOFT FOUNDATION

Practical Session

Practical Exercise 1

```
colors=("Red","Blue","Green")


print(colors)
print(len(colors))
```

Practical Exercise 2

```
numbers=(5,10,15,20)


for i in numbers:
    print(i)
```

Practical Exercise 3

```
fruits=("Apple","Banana","Mango")


print(fruits[2])
```

Practical Exercise 4

```
animals=("Dog","Cat","Tiger")


print("Cat" in animals)
```

Practical Exercise 5

```
marks=(90,80,70)
```

3DWEBSOFT FOUNDATION

```
name="Rahul"


print(name)
print(marks)
```

Mini Activity

Guess the Output

```
numbers=(10,20,30)


print(numbers[1])
```

Answer

20

Guess

```
colors=("Red","Blue")


print(colors*2)
```

Answer

('Red','Blue','Red','Blue')

Classroom Activity

Create a tuple of

- 5 Fruits
- 5 Animals
- 5 Colors

Print all values using a **for loop**.

3DWEBSOFT FOUNDATION

Quiz

Choose the Correct Answer

1. Tuple uses

A) {}

B) []

C) ()

D) <>

✅ Answer: **C**

2. Tuple is

A) Mutable

B) Immutable

C) Empty

D) Dictionary

✅ Answer: **B**

3. Which function gives tuple length?

A) size()

B) count()

C) len()

D) total()

✅ Answer: **C**

4. Can tuple contain duplicate values?

A) Yes

B) No

✅ Answer: **A**

3DWEBSOFT FOUNDATION

5. Which function finds an item's position?

A) search()

B) index()

C) locate()

D) find()

✅ Answer: **B**

Homework

1. Create a tuple of 5 favorite fruits.
2. Print the first and last fruit.
3. Find the length of the tuple.
4. Use a **for loop** to print all items.
5. Check whether "Apple" is present in the tuple.
6. Create a tuple of your family members' names.
7. Convert a list into a tuple.
8. Convert a tuple into a list.
9. Write a program using count().
10. Write a program using index().

Key Points to Remember

- A **Tuple** stores multiple values in one variable.
- Tuples use **round brackets ()**.
- Tuples are **ordered** and **immutable**.
- They can contain duplicate values.
- Use **indexing** to access items.
- Use **len()** to find the number of items.

3DWEBSOFT FOUNDATION

- Use **count()** and **index()** for common operations.
- Tuples are useful for storing data that should not change.

3DWEBSOFT FOUNDATION